Worksheet: Propositional Logic & First-Order Logic

Foundations of Artificial Intelligence – SS 2025 *University of Bremen, Institute for Artificial Intelligence*

> **Instructions:** Written tasks (✏) should be answered directly in this document or in a Markdown cell in the notebook. Programming tasks (▯) belong in `logic_exercises.ipynb`. Estimated time: **~60 minutes**.

Part 1 – Propositional Logic: Entailment & Satisfiability ✏ *(~10 min)*

1.1 Is the following entailment true or false? Justify your answer.

$$A \models A \lor B$$

1.2 Is the formula $(A \to B) \lor (B \to A)$ a tautology? Prove your answer using a truth table.

1.3 For each formula below, state whether it is **satisfiable**, **unsatisfiable**, or a **tautology**:

#	Formula	Satisfiable?	Unsatisfiable?	Tautology?
i	$A \land \neg A$			
ii	$A \lor \neg A$			
iii	$(A \to B) \land (B \to A)$			
iv	$\neg(A \lor B) \land A$			

1.4 Does $A \leftrightarrow B$ entail $A \to B$? Explain why.

1.5 Which of the following can be entailed from $A \land B$?

- (a) $A \lor B \lor C$
- (b) $A \leftrightarrow B$
- (c) None of the above

Part 2 – Counting Models ✏ *(~8 min)*

Consider a vocabulary with four propositional symbols: A, B, C, D.
There are $2^4 = 16$ possible truth assignments in total.

For each sentence, count how many of the 16 models satisfy it:

(a) $(A \wedge B) \vee (B \wedge C)$

(b) $A \vee B$

(c) $(A \leftrightarrow B) \leftrightarrow D$

Part 3 – Validity and Unsatisfiability ✏ *(~10 min)*

For each sentence, decide whether it is **valid** (tautology), **unsatisfiable**, or **neither**.
Briefly justify each answer — a truth table or a short argument is fine.

	Formula	Valid / Unsat / Neither
a	Smoke $\rightarrow$ Smoke	
b	$($Smoke $\rightarrow$ Fire$)$ $\rightarrow$ $(\neg$Smoke $\rightarrow$ $\neg$Fire$)$	
c	Smoke $\vee$ Fire $\vee$ $\neg$Fire	
d	$(($Smoke $\wedge$ Heat$)$ $\rightarrow$ Fire$)$ $\leftrightarrow$ $(($Smoke $\rightarrow$ Fire$)$ $\vee$ $($Heat $\rightarrow$ Fire$))$	
e	$($Smoke $\rightarrow$ Fire$)$ $\rightarrow$ $(($Smoke $\wedge$ Heat$)$ $\rightarrow$ Fire$)$	
f	$($Fire $\rightarrow$ Smoke$)$ $\wedge$ Fire $\wedge$ $\neg$Smoke	

Part 4 – Normal Forms ✏ *(~10 min)*

4.1 Convert the following expressions into **Conjunctive Normal Form (CNF)**:

(i) $\neg(s \to \neg p) \wedge (p \to \neg r)$

(ii) $p \leftrightarrow (\neg q \vee (q \to \neg p))$

(iii) $\neg(p \to (\neg q \vee (r \leftrightarrow \neg p)))$

4.2 Convert the following expression into **Disjunctive Normal Form (DNF)**:

$(s \to \neg(\neg r \to q)) \to p$

Part 5 – Python: Truth Tables *(~10 min)*

Open `logic_exercises.ipynb` and go to **Exercise 5**.

You are given a helper that evaluates a formula for one assignment. Your tasks:

5.1 Complete `truth_table(variables, formula_fn)` to print all rows.

5.2 Use it to verify your answers from Part 1.2 (the tautology) and Part 3 (validity table).

5.3 Use it to **count** the number of satisfying models for the formulas in Part 2.

Part 6 – Python: Resolution *(~8 min)*

In **Exercise 6** of the notebook you will implement a simple resolution checker.

6.1 A clause is a set of literals like `{'P', '¬Q', 'R'}`. Complete `resolve(c1, c2)` which returns the resolvent of two clauses (or `None` if they cannot be resolved).

6.2 Complete `pl_resolution(kb_clauses, goal_clause)` which tries to derive a contradiction from the KB + negated goal.

6.3 Use it to verify the unicorn problem from the worksheet: - If the unicorn is mystical → immortal; if not mystical → mortal mammal - Immortal or mammal → has horn; has horn → magical

Show that **the unicorn is magical**.

Part 7 – First-Order Logic: Formula Matching ✏ *(~5 min)*

Match each FOL formula to its correct English sentence. Write the letter next to the number.

#	Formula	Answer
1	$\exists x.\,(\textsf{big}(x) \wedge \textsf{cat}(x))$	
2	$\exists x.\,(\textsf{big}(x) \rightarrow \textsf{cat}(x))$	
3	$\exists x.\,(\textsf{big}(x) \vee \textsf{cat}(x))$	
4	$\forall x.\,(\textsf{big}(x) \wedge \textsf{cat}(x))$	
5	$\forall x.\,(\textsf{big}(x) \rightarrow \textsf{cat}(x))$	
6	$\forall x.\,(\textsf{big}(x) \vee \textsf{cat}(x))$	
7	$\forall x.\,(\textsf{big}(x) \vee \neg\textsf{cat}(x))$	

(A) All cats are big. **(B)** There exists a small thing or a cat. **(C)** All big things are cats. **(D)** All things are big cats. **(E)** All cats are small. **(F)** There exists a big cat. **(G)** All small things are cats. **(H)** There exists a big thing or a cat.

Part 8 – FOL Translation Critique ✏ *(~5 min)*

For each sentence, decide whether the FOL formula is a **correct** translation. If not, describe the mistake.

(a) *"Every apartment in Munich is cheaper than some apartment in Starnberg."*

$$\forall x.\,(\textsf{App}(x) \wedge \textsf{In}(x, \textsf{Munich}) \rightarrow \exists y.\,(\textsf{App}(y) \wedge \textsf{In}(x, \textsf{Starnberg}) \rightarrow \textsf{Rent}(x) < \textsf{Rent}(y)))$$

(b) *"There is exactly one apartment in Starnberg with rent below 500€."*

$$\exists x.\,(\textsf{App}(x) \wedge \textsf{In}(x, \textsf{Starnberg}) \wedge \forall y.\,(\textsf{App}(y) \wedge \textsf{In}(y, \textsf{Starnberg}) \wedge \textsf{Rent}(y) < 500 \rightarrow x = y))$$

(c) *"If some apartment is more expensive than any apartment in Munich, it must be in Starnberg."*

$$\forall x.\,(\mathsf{App}(x) \wedge \forall y.\,(\mathsf{App}(y) \wedge \mathsf{In}(y, \mathsf{Munich}) \wedge \mathsf{Rent}(x) > \mathsf{Rent}(y)) \rightarrow \mathsf{In}(x, \mathsf{Starnberg}))$$

Good luck! □